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Roll No: 46
BlockChain & DLT

EXPERIMENT NO - 8

Aim :- To implement the Blockchain platform Ganache and deploy a simple Solidity smart contract using Metamask and Remix IDE.

Theory :-

Q1. What is Blockchain?

Answer:
Blockchain is a decentralized and distributed digital ledger that records transactions securely across multiple nodes. It ensures transparency, immutability, and security, making it suitable for applications where trust and data integrity are important.

Q2. What is Ganache?

Answer:
Ganache is a personal blockchain used for Ethereum development and testing. It provides a local blockchain environment with pre-configured accounts, allowing developers to deploy, test, and debug smart contracts without using real cryptocurrency.

Q3. What is the role of MetaMask in this experiment?

Answer:
MetaMask is a digital wallet used to connect the local Ganache blockchain with Remix IDE. It allows users to manage accounts, sign transactions, and interact with deployed smart contracts.

Q4. How is Blockchain used in charity donation systems?

Answer:
Blockchain enables transparent and traceable donation systems. Every transaction is recorded on the blockchain, allowing donors to track how their funds are used. This increases trust and reduces the chances of fraud or misuse.

Q5. What is the role of Smart Contracts in charity donations?

Answer:
Smart contracts automate the donation process by executing predefined conditions. For example, funds can be released only when certain conditions are met. This ensures secure, transparent, and efficient handling of donations without intermediaries.

Q6. Why is Blockchain beneficial for charity systems?

Answer:
Blockchain provides transparency, security, and accountability. Donors can verify transactions, ensuring that funds reach the intended recipients. It also reduces administrative costs and eliminates the need for intermediaries.

Q7. What are the steps involved in connecting Ganache with MetaMask and Remix IDE?

Answer:
The steps include installing Ganache, importing Ganache accounts into MetaMask, connecting MetaMask to the local Ganache network, and linking it with Remix IDE for deploying and interacting with smart contracts.

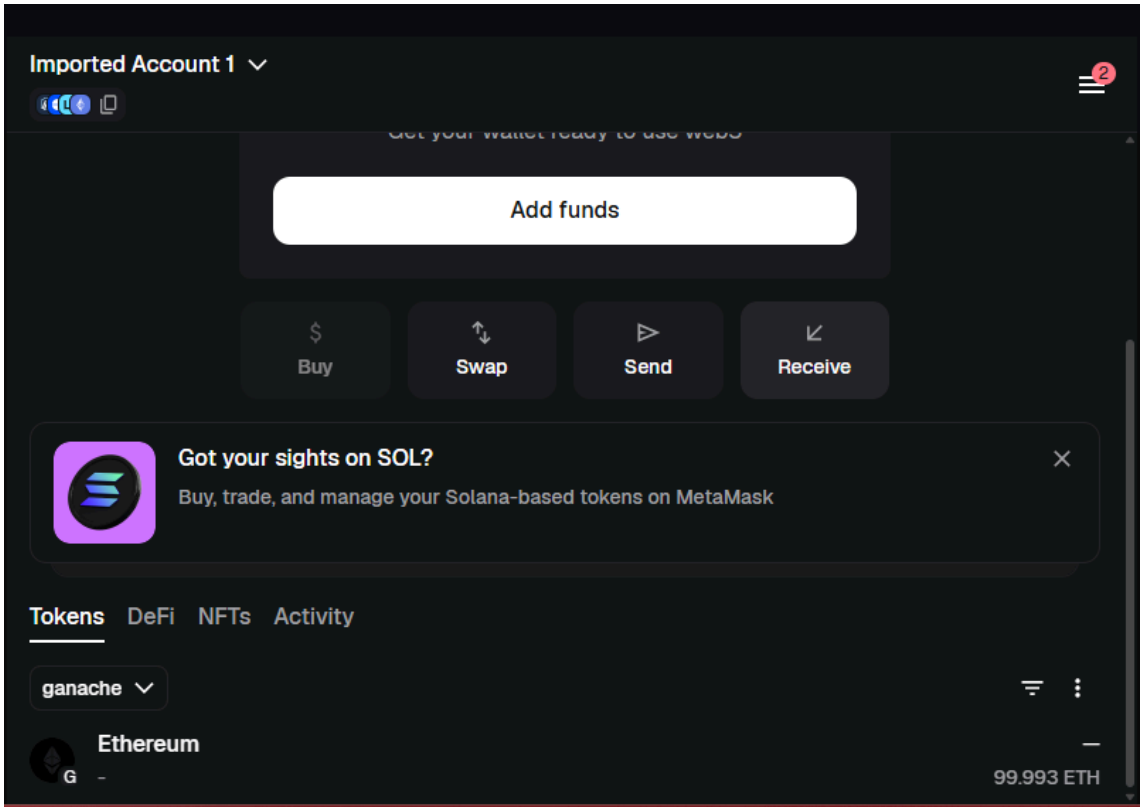
Implementation :-

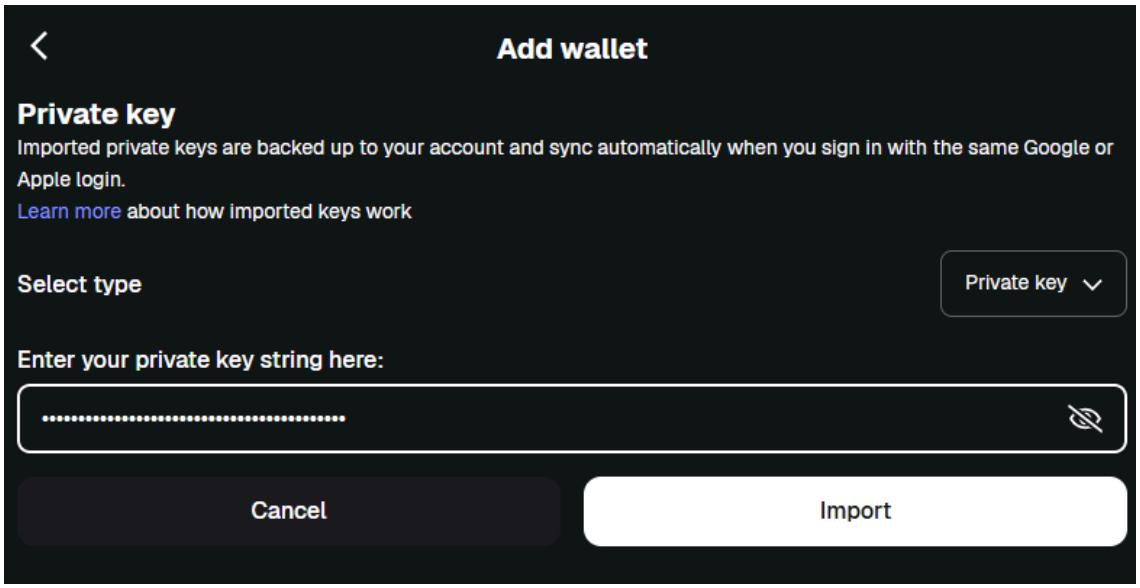
```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.0;

contract CharityDonation {

    address public owner;
    uint public totalDonations;

    struct Donation {
        address donor;
        uint amount;
        uint timestamp;
    }

    Donation[] public donations;

    event DonationReceived(address donor, uint amount);
    event FundsWithdrawn(address owner, uint amount);

    constructor() {
        owner = msg.sender;
    }
```

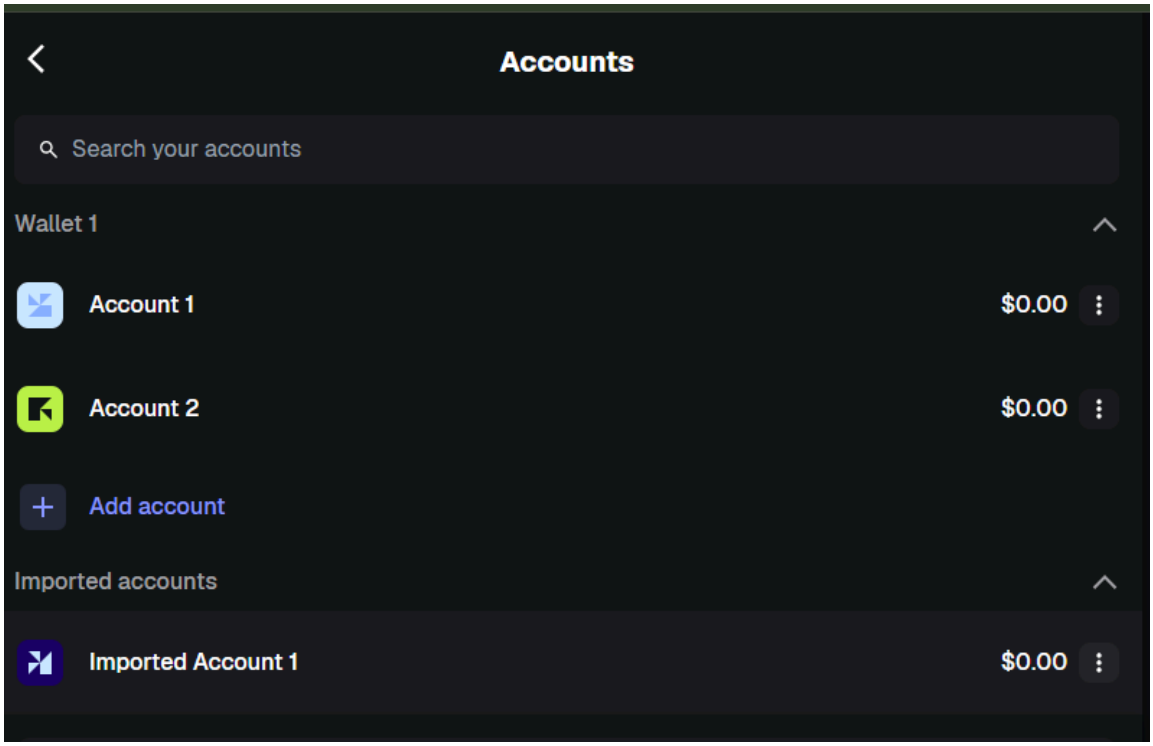
```
    // Function to donate Ether
    function donate() public payable {
        require(msg.value > 0, "Donation must be greater than zero");

        donations.push(Donation(msg.sender, msg.value, block.timestamp));
        totalDonations += msg.value;

        emit DonationReceived(msg.sender, msg.value);
    }

    // Function to check contract balance
    function getBalance() public view returns (uint) {
        return address(this).balance;
    }

    // Only owner can withdraw funds
    function withdraw(uint amount) public {
        require(msg.sender == owner, "Only owner can withdraw funds");
        require(amount <= address(this).balance, "Insufficient balance");

        payable(owner).transfer(amount);

        emit FundsWithdrawn(owner, amount);
    }

    // View number of donations
    function getDonationCount() public view returns (uint) {
        return donations.length;
    }
}
```

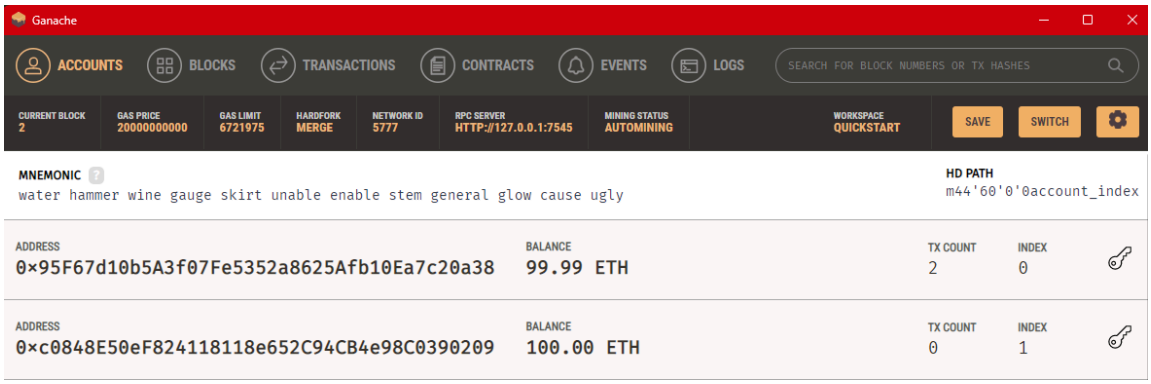
Metamask

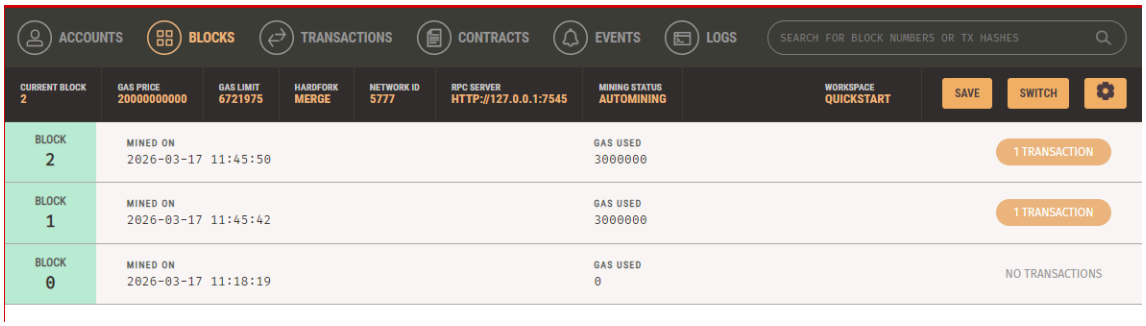
Output:


Imported Account 1
2
Add funds
Buy
Swap
Send
Receive
Got your sights on SOL?
Buy, trade, and manage your Solana-based tokens on MetaMask
Tokens
DeFi
NFTs
Activity
ganache
Ethereum
G
99.993 ETH


Add wallet
Private key
Imported private keys are backed up to your account and sync automatically when you sign in with the same Google or Apple login.
Learn more about how imported keys work
Select type
Private key
Enter your private key string here:
Cancel
Import


Accounts
Search your accounts
Wallet 1
Account 1
$0.00
Account 2
$0.00
Add account
Imported accounts
Imported Account 1
$0.00


Ganache
ACCOUNTS
BLOCKS
TRANSACTIONS
CONTRACTS
EVENTS
LOGS
SEARCH FOR BLOCK NUMBERS OR TX HASHES
CURRENT BLOCK 2
GAS PRICE 20000000000
GAS LIMIT 6721975
HARDFORK MERGE
NETWORK ID 5777
RPC SERVER HTTP://127.0.0.1:7545
MINING STATUS AUTOMINING
WORKSPACE QUICKSTART
SAVE
SWITCH
MNEMONIC
water hammer wine gauge skirt unable enable stem general glow cause ugly
HD PATH
m44'60'0'0account_index
ADDRESS
0x95F67d10b5A3f07Fe5352a8625Afb10Ea7c20a38
BALANCE
99.99 ETH
TX COUNT
2
INDEX
0
ADDRESS
0xc0848E50eF824118118e652C94CB4e98C0390209
BALANCE
100.00 ETH
TX COUNT
0
INDEX
1


ACCOUNTS
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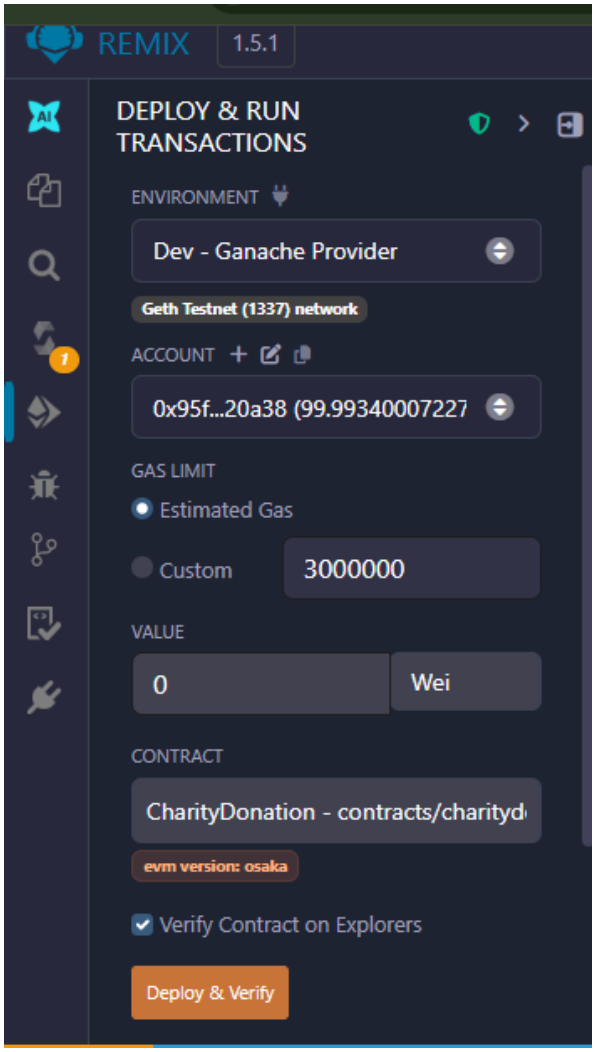
HARDFORK MERGE
NETWORK ID 5777
RPC SERVER HTTP://127.0.0.1:7545
MINING STATUS AUTOMINING
WORKSPACE QUICKSTART
SAVE
SWITCH
BLOCK 2
MINED ON 2026-03-17 11:45:50
GAS USED 3000000
1 TRANSACTION
BLOCK 1
MINED ON 2026-03-17 11:45:42
GAS USED 3000000
1 TRANSACTION
BLOCK 0
MINED ON 2026-03-17 11:18:19
GAS USED 0
NO TRANSACTIONS



Conclusion :-

This experiment demonstrated the implementation of a blockchain-based environment using Ganache, MetaMask, and Remix IDE. A smart contract was successfully created, deployed, and tested using Ganache accounts integrated with MetaMask. The experiment also highlighted the potential of blockchain technology in enhancing charity donation systems by ensuring transparency, security, and traceability of transactions. Overall, it provided practical knowledge of executing and monitoring blockchain transactions in a controlled environment.